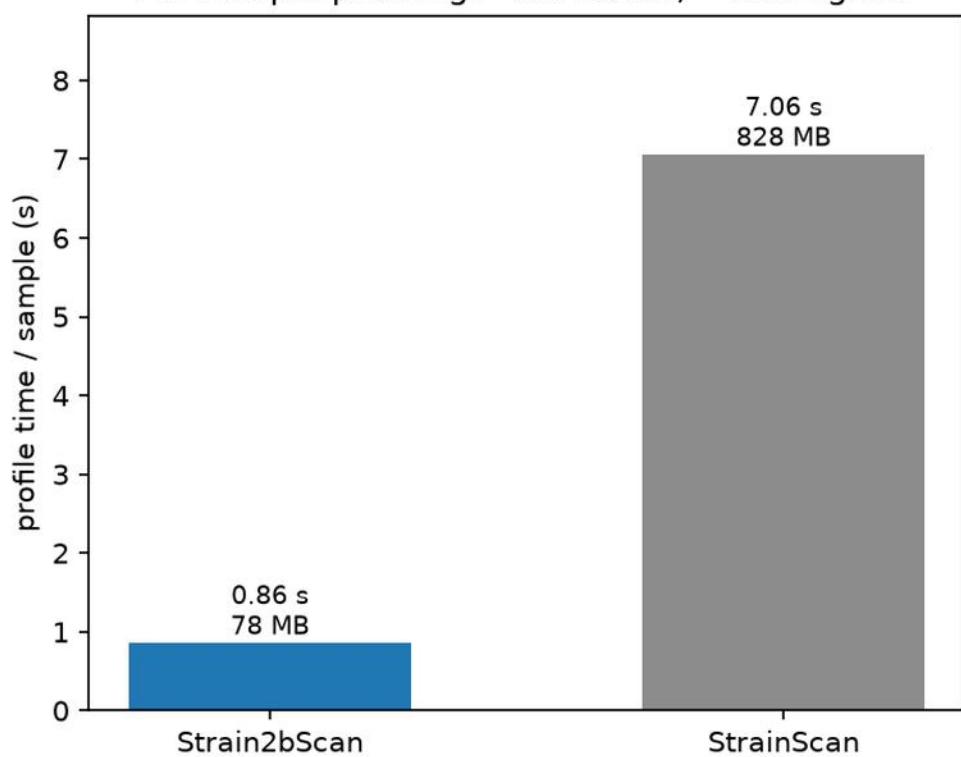
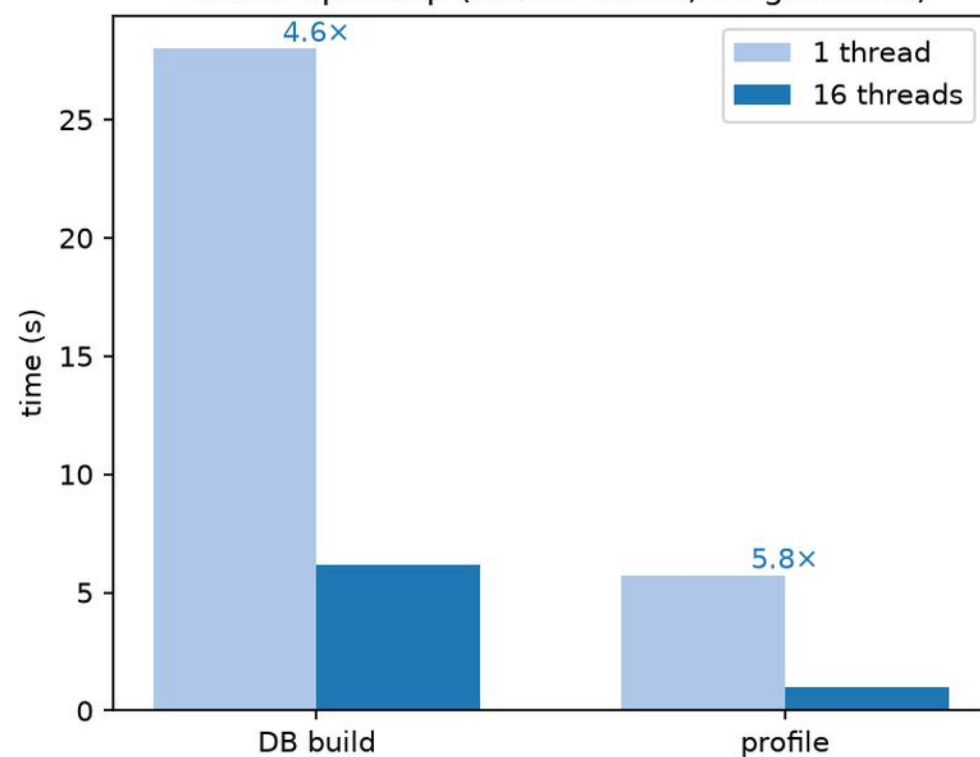
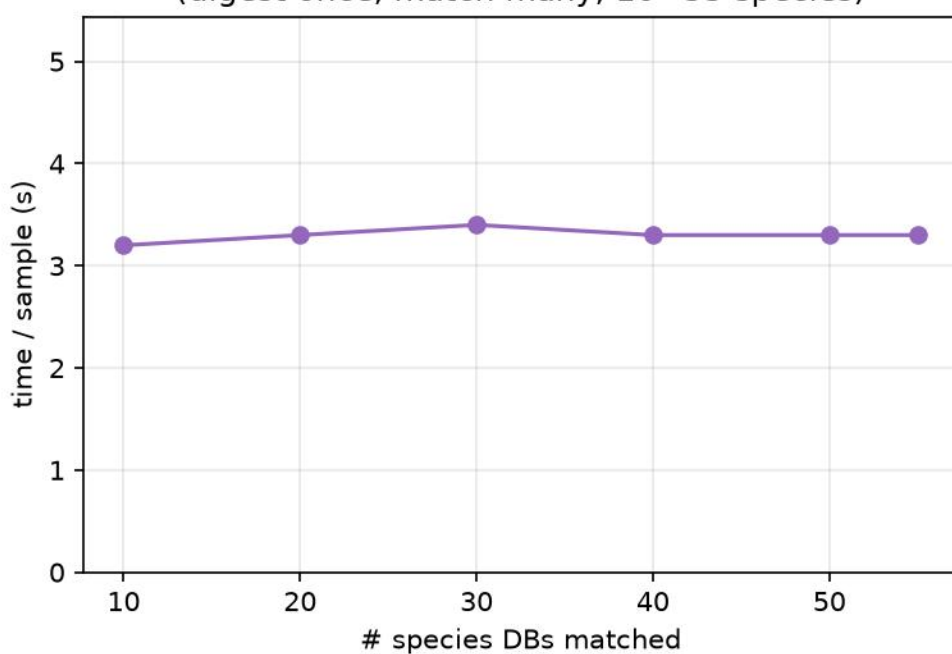
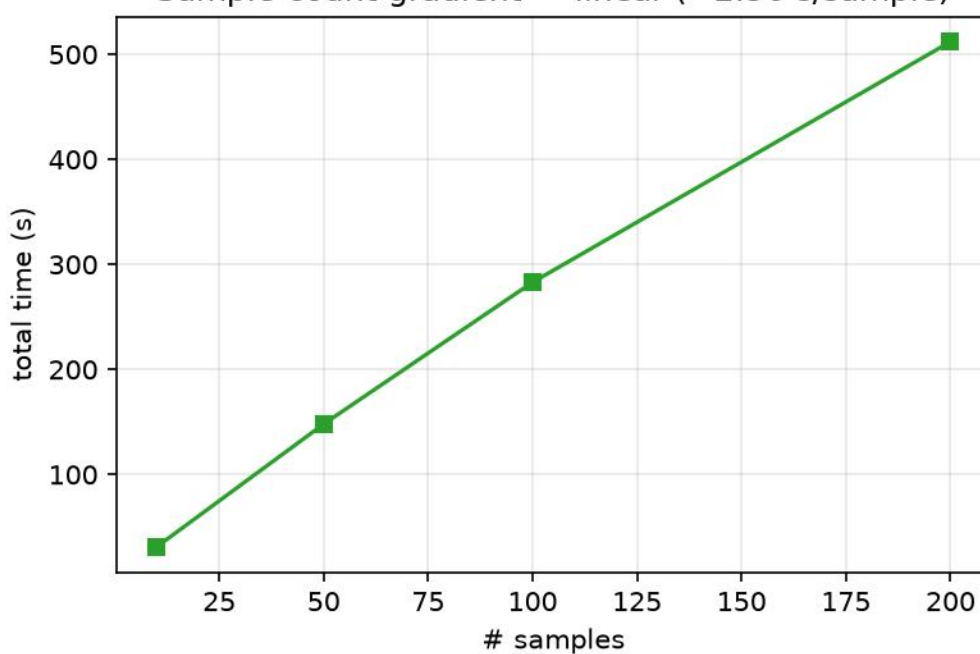


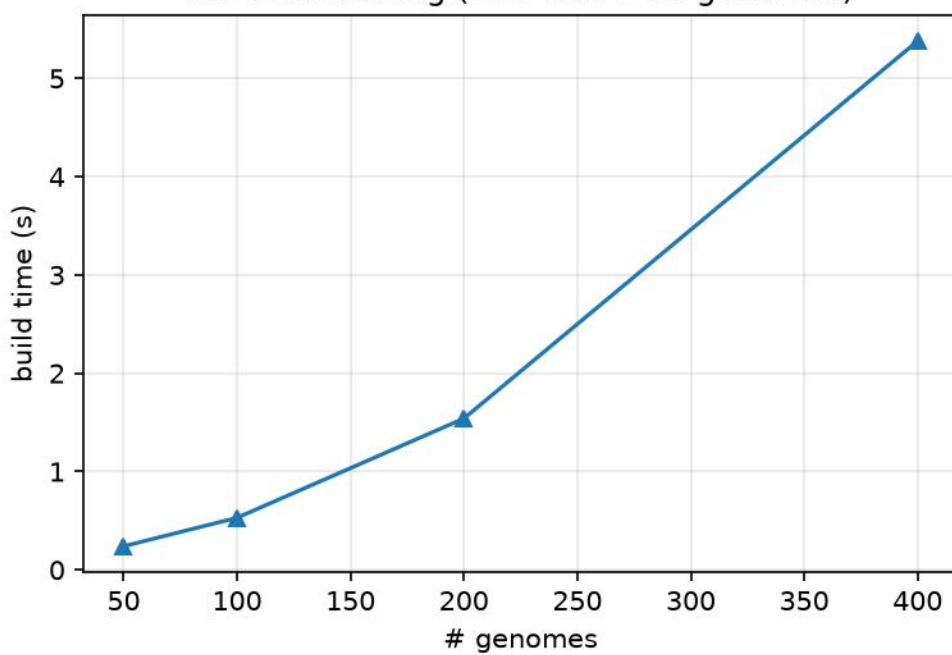
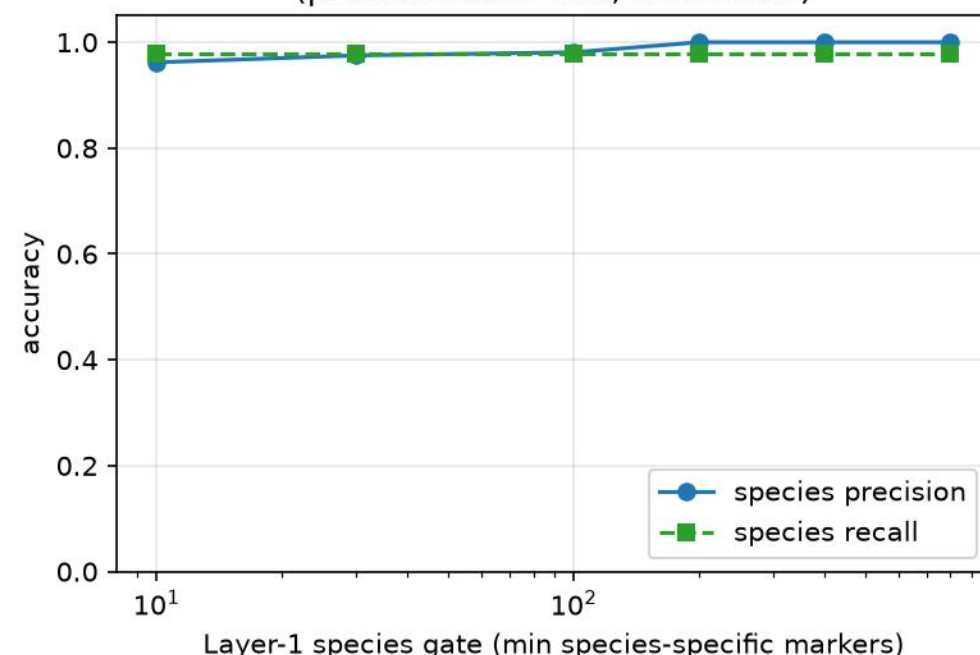
Per-sample profiling: ~8× faster, ~11× lighter

Parallel speedup (real *C. acnes*, 64 genomes)Species-count gradient — flat
(digest once, match many; 10→55 species)

Sample-count gradient — linear (~2.56 s/sample)



DB-build scaling (MinHash >96 genomes)

Multi-species accuracy vs species gate
(precision 0.96→1.0, recall 0.98)

Community-scale strain profiling (55 real species)

